Yulong Liu

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Research Interests

Computational Geomechanics | Poromechanics | Reservoir Simulation | Scientific Machine Learning

Education

Ph.D. in Earth Science – Cornell University 2024 – Present
Minors: Computer Science and Scientific Computing
B.S. in Mining Engineering – Northeastern University (CN) 2024

Research Experience

Ph.D. Candidate – Dept. of Earth and Atmospheric Sciences, Cornell University Aug. 2024 – Present
Advisor: Chloé Arson *Ithaca, NY*
Project: Computational geomechanics and poromechanics for coupled multiphysics problems in subsurface and reservoir systems, supported by physics-based and machine-learning surrogate modeling

- Develop coupled poromechanics and computational geomechanics models of subsurface and reservoir systems (hydraulic fracturing, geothermal injection–production) in the MOOSE finite element framework, capturing thermo-hydro-mechanical interactions in fractured rock.
- Study multiscale transport in porous subsurface media through homogenization theory, informing reservoir-scale surrogate modeling.
- Build physics-informed neural networks, implicit neural representations, and operator-learning surrogates that deliver geometry- and physics-aware predictions at a fraction of the cost of full finite element simulation, supporting engineering decision-making.
- Lead a cross-institution project on PDE-grounded verification of the physical correctness of LLM-generated multiphysics simulation code.
- **Ongoing:** AI-driven discovery of closed-form homogenization laws for permeability—combining data-driven methods with homogenization theory to learn interpretable closed-form expressions for upscaled permeability of heterogeneous porous media.

Undergraduate Researcher – Northeastern University (CN) Sep. 2020 – Jun. 2024
B.S. in Mining Engineering *Shenyang, China*
Thesis: A Unified Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning

- Developed a unified deep-learning model for microseismic signal classification and arrival-time picking, integrating tasks typically handled in separate workflows.
- Conducted experimental and computational research on hydraulic fracturing, triaxial rock testing, and tunnel boring machines; received the Outstanding Thesis Award and National Scholarship.

Grants, Proposals, and Computational Awards

Learned Uncertainty-Propagation and World Models for Enhanced Geothermal Reservoirs 2026
National Science Foundation (NSF), ACCESS program. Student lead: Y. Liu; Advisor: C. Arson. Total allocation and Arson group's share: 1.5M CPU hours.
AI-aided Computational Geomechanics 2026
Empire AI Alpha+Beta. PIs: Y.C. Han, Y. Liu, A. Tristani, and C. Arson. Total allocation and Arson group's share: 5.5K GPU hours.
NASA FINESST Proposal – Title Withheld During Anonymous Review 2026
NASA ROSES-25 F.5 FINESST. Under review. Future Investigator: Y. Liu; PI and faculty mentor: C. Arson.

Publications

Selected Papers

- **Yulong Liu**, Jonah Botvinick-Greenhouse, Yunan Yang, Chloé Arson. *Operator Learning Surrogate Modeling of Hydraulically Fractured Geothermal Injection–Production Systems: A Cornell Case Study.* ARMA US Rock Mechanics/Geomechanics Symposium, Tucson, AZ, 2026.
- **Liu, Y.**, Arson, C. *A Physics-Informed Neural Network for Pressurized Cavities of Arbitrary Shape in Heterogeneous Rock. Rock Mech Rock Eng* (2026). doi:10.1007/s00603-026-05882-5.
- M. Belachew, **Yulong Liu**, J. D. Frost, Chloé Arson. *Numerical Assessment of Plasticity Development and Energy Expenditure of Ant-Like Microtunnelling. Tunnelling and Underground Space Technology*, 172, 107501, 2026.
- **Yulong Liu**†, Chloé Arson. *Physics-Informed Neural Network Surrogate Modeling of Pressurized Cavity in Homogeneous and Bilayered Media.* ARMA US Rock Mechanics/Geomechanics Symposium, D022S018R006, 2025.
- **Yulong Liu***,†, Zhenghan Song, et al. *Your Simulation Runs but Solves the Wrong Physics: PDE-Grounded Intent Verification for LLM-Generated Multiphysics Simulation Code.* Under review at NeurIPS 2026.
- Zhenghan Song, Yunyi Li, **Yulong Liu**†. *Prefix-Safe Bayesian Belief Tracking for LLM Reasoning Reliability: Separating Calibration from Ranking.* Under review at ACL ARR 2026.

Thesis

- **Yulong Liu**. *A Unified Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning.* Undergraduate thesis, Northeastern University, 2024.

* Co-first author † Corresponding author

Conferences and Presentations

Oral presentation, 60th US Rock Mechanics/Geomechanics Symposium (ARMA) Jun. 2026
American Rock Mechanics Association, Tucson, AZ, USA

Invited talk, 20th Phase Field Methods Workshop May 2026
Center for Hierarchical Materials Design, Northwestern University, Evanston, IL, USA

Poster presentation, Cornell CEE Graduate Research Symposium Apr. 2026
Cornell University, Ithaca, NY, USA

Invited seminar: "Physics-Informed AI Optimization of Heterogeneous Rock Systems" Dec. 2025
Southwest Petroleum University seminar series, Chengdu, China

Poster presentation, 59th US Rock Mechanics/Geomechanics Symposium (ARMA) Jun. 2025
American Rock Mechanics Association, Santa Fe, NM, USA

Fellowships and Awards

Estwing Hammer Award – Most Outstanding EAS Graduate Student of 2024, Cornell University	2025
Cornell University Travel Grant, Cornell University	2025, 2026
Outstanding Thesis, Northeastern University	2024
National Scholarship, Ministry of Education, China	2024
First Class Scholarship, Northeastern University	2023
Autumn Scholarship, Northeastern University	2022
Golden Seed Scholarship, Northeastern University	2022

Academic Service

- Reviewer, *Rock Mechanics and Rock Engineering*
- Reviewer, *Underground Space*
- Reviewer, *International Conference on Artificial Neural Networks*
- Reviewer, *Geothermal Rising Conference*

Skills

Programming & Scientific Computing: Python, C++, R, MOOSE finite element framework **Modeling & Simulation:** finite element modeling, coupled multiphysics simulation, poromechanics, computational geomechanics, CAD (e.g., AutoCAD, SolidWorks) **Subsurface & Reservoir:** hydraulic fracturing, geothermal injection–production modeling, multiscale transport in porous media **Scientific Machine Learning:** ML for PDEs, physics-informed neural networks, operator learning, implicit neural representations (INRs), surrogate modeling for scientific computing **LLMs for Simulation:** using large language models to accelerate simulation-software workflows and to verify the physical correctness of generated simulation code **Experimental Methods:** rock mechanics testing (e.g., triaxial compression), laboratory instrumentation **Currently Learning:** phase-field modeling (fracture/multiphysics), multiphase flow in porous media **Scientific Writing:** LaTeX / Overleaf

Professional Memberships

- Member, American Rock Mechanics Association (ARMA), 2025–Present
- Member, International Society for Rock Mechanics and Rock Engineering (ISRM), 2025–Present
- Member, Chinese Society for Rock Mechanics and Engineering (CSRME), 2024–Present
- Member, Geothermal Rising, 2025–Present